

NOTE: .csv tables are included showing these results more clearly where stated in capstone project notes

Docker compose showing container started

```
• (venv) MAC@MacBook-Air week-7-capstone % docker compose -f docker-compose.cost.yml up -d
[+] up 2/2
  ✓ Network week-7-capstone_default          Created      0.0s
  ✓ Container week-7-capstone-triage-api-1    Started      0.1s
○ (venv) MAC@MacBook-Air week-7-capstone %
```

Curl <http://localhost:8000/health> shows API responding

Docker inspect shows FinOps labels on the running container

```
• (venv) MAC@MacBook-Air week-7-capstone % docker compose -f docker-compose.cost.yml ps -a
NAME                                IMAGE                                COMMAND                                SERVICE    CREATED        STATUS                PORTS
week-7-capstone-triage-api-1        triage-api:1.0.0                    "uvicorn triage_mode..."            triage-api  4 minutes ago  Up 4 minutes (healthy)  0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp
• (venv) MAC@MacBook-Air week-7-capstone % curl http://localhost:8000/health
{"service":"triage-api","version":"1.0.0","status":"healthy"}
• (venv) MAC@MacBook-Air week-7-capstone % docker inspect --format '{{json .Config.Labels}}' $(docker c
compose -f docker-compose.cost.yml ps -q triage-api)
{"afyapplus.cost-center":"clinical-ai","afyapplus.service":"triage-api","afyapplus.version":"1.0.0","com.docker.compose.config-hash":"63fb2966abba945ce28f2d425ea6951f09bcf4620cea6af65f3390897da56720","com.docker.compose.container-number":"1","com.docker.compose.depends_on":"","com.docker.compose.image":"sha256:988418dad983699026e1f3dba8b20973d15f4dbc0a290046916ece4a954a3d1a","com.docker.compose.oneoff":"False","com.docker.compose.project":"week-7-capstone","com.docker.compose.project.config_files":"/Users/MAC/afyapplus-platform/week-7-capstone/docker-compose.cost.yml","com.docker.compose.project.working_dir":"/Users/MAC/afyapplus-platform/week-7-capstone","com.docker.compose.service":"triage-api","com.docker.compose.version":"5.5.1"}
```

Checked run on azure_budget_triage.json

```
● (venv) MAC@MacBook-Air week-7-capstone % python -c "import json; d=json.load(open('azure_budget_triage.json')); \n\n    print(d['filter'], d['notifications']['Actual_GreaterThan_80_Percent']['threshold'])"\n\n{'tags': {'service': ['triage-api']}} 80
```

Checked run on aws_budget.json

```
● (venv) MAC@MacBook-Air week-7-capstone % python -c "import json; d=json.load(open('aws_budget.json'));\n\n    print(d['BudgetLimit'], d['CostFilters'])"\n\n{'Amount': '400', 'Unit': 'USD'} {'TagKeyValue': ['user:service$triage-api']}
```

At least two levers implemented

```
• (venv) MAC@MacBook-Air week-7-capstone % python cache_hitrate.py
  backend=memory hits=60 misses=40 hit_rate=0.60
• (venv) MAC@MacBook-Air week-7-capstone % python batch_worker.py
  processed 3 model calls 1
```

Before/after metrics: including hit-rate and/or \$/1k and/or p95 on a stated fixture or load script

```
• (venv) MAC@MacBook-Air week-7-capstone % python levers_bench.py
lever scenario      calls  hit  $/1k  p95 ms  backend
cache before (no cache)    100  0.0  0.2018  0.00  memory/stub
cache after (cache on)     40  0.6  0.0807  0.00  memory/stub
batch before (one call per job) 100  0.1052 0.75  sqlite/stub
batch after (batch of 8)    13   0.0536 4.05  sqlite/stub
cache saving on $/1k: 60%
batch saving on $/1k: 49%

wrote results/levers_before_after.csv

[QUALITY]
hit answers identical to original miss: 60/60
hits after prompt-version bump: 0 (expected 0)
urgency accuracy: SKIPPED (stub backend answers 'routine' for everything; run with USE_LLM=1)

NOTE: stub backend. Call counts, hit rate and $/1k are valid; p95 reflects local code only, not model latency.
```

Breakeven sketch with assumption

```
(venv) MAC@MacBook-Air week-7-capstone % python breakeven.py
api_usd_per_req=0.000202
selfhost_fixed_monthly=1300.00 (ESTIMATE)
selfhost_var_per_req=0.000050 (ESTIMATE)
```

scenario	break-even/month	x baseline
Base case	8.6M	85.6x
API price -50%	25.5M	255.3x
API price +50%	5.1M	51.4x
GPU -40% (reserved/spot)	6.3M	63.2x